

Time Series Analysis 2026

Capstone Report | IIT Guwahati x StockGro

1. Methodology and Models Used

We applied time series forecasting to 11 NSE stocks, allocated Rs.10,00,000 in a virtual portfolio on StockGro, and compared predictions against actual market outcomes.

Dataset: Yahoo Finance (yfinance), Jan 2021 - Dec 2025, daily. Split: Training (< Jul 2025) | Testing (Jul-Dec 2025, 125 obs).

Model	Library	Purpose
ARIMA/SARIMA	pmdarima	Price forecasting (auto order via AIC)
Facebook Prophet	prophet	Trend + seasonality with Indian holidays
LSTM (2-layer)	PyTorch	Non-linear deep learning sequence model
Exp. Smoothing	statsmodels	Holt-Winters triple exponential smoothing
GARCH(1,1)	arch	Conditional volatility estimation

2. Stock Selection Rationale (Task 1)

We screened 25 candidate stocks across 5 sectors using three quantitative criteria: (1) 30-day rolling std dev, (2) STL trend strength, (3) 6-month sector momentum. Selected 11 stocks:

Sector	Selected Stocks	Key Justification
Banking	HDFC Bank, ICICI Bank, SBI	Highest liquidity (15-25M daily vol)
IT	TCS, Infosys	Top-2 by market cap, mean-reversion
Pharma	Sun Pharma, Dr. Reddy's	Low correlation, defensive sector
FMCG	ITC, Hindustan Unilever	Lowest volatility (4th-8th pctl)
Auto	M&M, Maruti	Highest sector momentum (+25.6%)

14 candidates rejected due to lower liquidity, weaker trends, or sector redundancy.

3. Data Preprocessing Steps (Task 2)

- Missing values: Forward-fill for market holidays and trading halts
- Stationarity:

4. Forecast Results with Confidence Intervals (Task 3)

4 models trained on each stock, validated on 125 test points, forecasting 2 days with 95% CIs:

Model	Avg MAPE	Best Stock	Worst Stock	Avg Dir. Acc.
ARIMA	0.77%	HDFCBANK (0.55%)	M&M (1.01%)	50.6%
LSTM	1.39%	HDFCBANK (0.69%)	M&M (2.87%)	47.9%
ExpSmoothing	6.50%	HDFCBANK (2.47%)	MARUTI (14.84%)	48.9%
Prophet	7.64%	DRREDDY (2.73%)	MARUTI (17.33%)	52.9%

ARIMA achieved the best accuracy with sub-1% average MAPE across all 11 stocks.

5. Volatility and Trend Analysis (Task 4)

GARCH(1,1) Volatility

High-persistence stocks (DRREDDY: 0.997, SUNPHARMA: 0.995) show slowly-decaying volatility clustering. Low-persistence stocks (INFY: 0.174, HINDUNILVR: 0.563) have mean-reverting volatility.

STL Trend Analysis

7 stocks upward (SBIN +15.76%, M&M +9.25%, MARUTI +9.08%), 2 sideways (ITC, DRREDDY), 2 downward (ICICIBANK -2.89%, HUL -4.33%).

6. Portfolio Composition and Allocation (Task 5)

Blended 4 strategies: Forecast-Guided (40%), Volatility-Aware (30%), Correlation-Diversified (20%), Sector Momentum (10%).

Stock	Weight	Amount (Rs.)	Primary Driver
ICICI Bank	13.5%	1,34,614	Forecast-Guided
Maruti	11.9%	1,19,072	Sector Momentum
M&M	11.0%	1,10,239	Sector Momentum
HDFC Bank	8.2%	81,897	Volatility-Aware
Sun Pharma	8.2%	81,869	Volatility-Aware
Dr. Reddy's	8.1%	81,172	Correlation-Diversified
HUL	8.1%	80,646	Volatility-Aware
ITC	8.0%	80,132	Volatility-Aware
TCS	7.9%	78,751	Correlation-Diversified
SBI	7.7%	76,708	Forecast-Guided
Infosys	7.5%	74,900	Correlation-Diversified
TOTAL	100%	10,00,000	

7. Model Comparison (Task 6)

Model	Avg MAPE	Avg RMSE	Dir. Acc.	Assessment
ARIMA	0.77%	33.7	50.6%	Best accuracy; interpretable
LSTM	1.39%	62.3	47.9%	Good non-linear; slow
ExpSmooth	6.50%	362.6	48.9%	Simple; poor on volatile
Prophet	7.64%	406.4	52.9%	Best dir. acc.; overestimates

ARIMA was used for portfolio allocation (Strategy A) due to lowest MAPE. GARCH used for volatility estimation (Strategy B).

8. StockGro Execution Summary (Task 7)

Trading Window: Day 1 = May 12, 2026 | Day 2 = May 13, 2026. All 11 stocks purchased on Day 1 at market open. Held without changes through Day 2. Total invested: Rs.9,89,803.

9. Predicted vs Actual Outcomes (Task 8)

Stock	Pred D1	Act D1	Err D1	Pred D2	Act D2	Err D2
SBIN	973.52	961.70	1.23%	974.10	956.52	1.84%
DRREDDY	1265.61	1257.58	0.64%	1265.61	1254.76	0.86%
HDFCBANK	990.90	739.79	33.94%	990.90	743.37	33.30%
HUL	2290.20	2223.38	3.01%	2290.20	2237.27	2.37%

ITC	392.56	298.29	31.60%	392.75	302.30	29.92%
SUNPHARMA	1710.40	1834.73	6.78%	1709.43	1810.54	5.58%
INFY	1621.60	1126.62	43.93%	1621.60	1115.42	45.38%
ICICIBANK	1343.85	1227.15	9.51%	1346.29	1215.45	10.76%
M&M	3662.70	3154.30	16.12%	3665.10	3091.93	18.54%
MARUTI	16663.28	12910.20	29.07%	16670.25	12927.68	28.95%
TCS	3188.83	2261.28	41.02%	3188.83	2234.92	42.68%
AVG MAPE			19.71%			20.02%

Portfolio: Invested Rs.9,89,803 -> Day 2 value Rs.9,69,134 -> Return: -2.07% (loss Rs.20,669)

Directional Accuracy: 6/11 (54.5%)

Analysis: Stocks like HDFCBANK, ITC, INFY, TCS show 30-45% errors due to likely corporate actions (stock splits) between Jan-May 2026. Stocks near Dec 2025 levels (SBIN: 1.23%, DRREDDY: 0.64%) had excellent predictions.

10. Reflections

What Worked

- ARIMA delivered excellent backtest accuracy (0.77% avg MAPE)
- Data-driven s

What Didn't Work

- Backtest vs live gap: 0.77% -> 20% MAPE (5-month extrapolation gap)
- Corporate ac

What I Would Improve

- 1. Retrain models daily on latest data before live predictions
- 2. Include ex